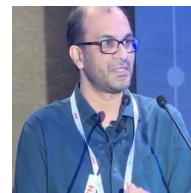


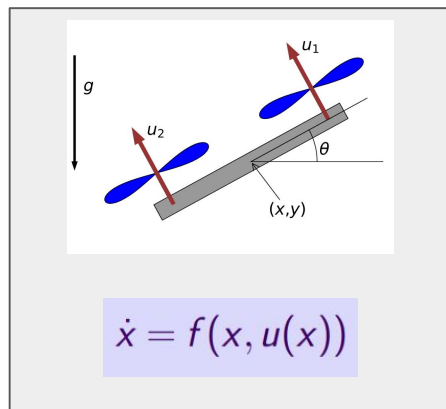
Blending Neural Control Density Functions for Stabilization and Safety

Sahil Chaudhary*, Chaitanya Murti*, and Chiranjib Bhattacharyya



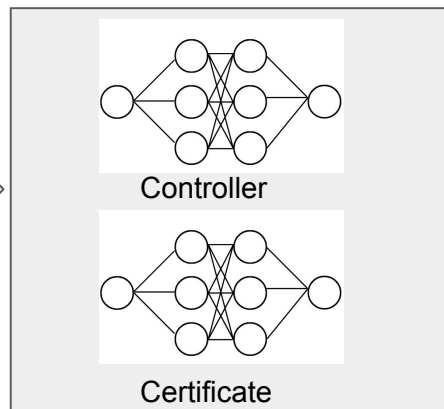
Synthesizing Stable Neural Network Controllers

Dynamical System (Known)



Training

Synthesizing Neural Network
Controller with certificate



Neural Lyapunov Control (**Chang et al. 2019**) [NeurIPS]

Lyapunov-stable neural-network control (**Dai et al. 2021**)

Neural lyapunov control for discrete-time systems (**Wu et al. 2023**) [NeurIPS]

Lyapunov-stable Neural Control for State and Output Feedback: A Novel Formulation (**Yang et al. 2024**) [ICML]

Two-Stage Learning of Stabilizing Neural Controllers via Zubov Sampling and Iterative Domain Expansion (**Li et. al 2025**) [NeurIPS]

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Lyapunov functions are a cornerstone of control theory, **but have major drawbacks**

- Fail to certify stability when **multiple equilibria** are in the region of interest
- Cannot be used to **blend controllers with a stability guarantee** - set of controller/certificate pairs is in general **not even connected!**

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In this work,

- We propose a **novel exponential parameterization** to learn Neural Control Density Functions (NCDFs) which certifies almost everywhere stability.

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- We propose a **novel exponential parameterization** to learn Neural Control Density Functions (NCDFs) which certifies almost everywhere stability.
- Characterize almost everywhere region of attraction(RoA) certified by density functions and show the blended controllers using density functions improves RoA.
- We extend our framework to **synthesize safe and stable controllers** via density and barrier functions.

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Theorem 1. [Rantzer 2001] *Given dynamical system $\dot{x} = f(x)$, 0 is an almost everywhere stable equilibrium if there exists $\rho(x) > 0$ s.t.*

$$\int_{\|x\|>1} \rho(x) f(x) / \|x\| dx < \infty \quad \text{and} \quad \nabla \cdot [f\rho](x) > 0$$

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Density functions also possess remarkable convexity property -> **convex nonlinear stabilization!**

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***Almost everywhere* Regions of Attraction (ROAs)** can be approximated with density functions!

Theorem 2. *Given dynamical system $\dot{x} = f(x)$, and density function ρ , D is an almost everywhere stable with ROA*

$D := \{x : \rho(x) \geq \tau\}$ if

$$\nabla \cdot [f\rho](x) > \tau \nabla \cdot [f](x) \quad \text{for all } x \in \partial D$$

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Given systems

$$\dot{x} = F_i(x) = f(x) + g(x)u_i(x), \quad i \in [N].$$

and density functions ρ_i that certify the a.e. stability of F_i . Define

$$\rho^*(x) = \sum_i \rho_i(x), \quad u^*(x) = \sum_i \frac{\rho_i(x)u_i(x)}{\rho^*(x)},$$

$$F^*(x) = f(x) + g(x)u^*(x),$$

Blending always improves Regions of Attraction!

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Theorem 3. *Suppose we have vector fields F_i with density functions ρ_i , with $\{x : \rho_i(x) > \tau\} \equiv D_i \subset S$. Let F^* , ρ^* and u^* be blended vector field, density, and controller, with $D^* := \{x : \rho^*(x) \geq \tau\}$ and let Q^* be the true almost-everywhere region of attraction. Then,*

$$\bigcup_i D_i \subseteq D^* \subseteq Q^*.$$

Neural Control Density Functions

The divergence condition: $\nabla \cdot [\rho(x)f(x) + g(x)u(x)] > 0$ is convex in $(\rho, \rho u)$!

$$\psi(x) = \rho(x)u(x)$$

$$\rho(x) = \frac{a(x)}{e^{\|x\|^2 + \|b(x)\|^2}}$$
$$\psi(x) = \frac{c(x)}{e^{\|x\|^2 + \|b(x)\|^2}}$$

Under mild assumption on the growth of function 'a' and 'c', such parameterization satisfy the **integrability condition**!

Neural Control Density Functions

- Represent 'a', 'b' and 'c' using Neural Networks and define the loss as,

$$L(\theta) = \mathbb{E}[\max(0, -\rho_\theta(x)) + \max(0, -\nabla \cdot [\rho_\theta(x)(f(x) + g(x)u_\theta(x))])]$$

$$\rho(x) = \frac{a(x)}{e(\|x\|^2 + \|b(x)\|^2)}$$
$$\psi(x) = \frac{c(x)}{e(\|x\|^2 + \|b(x)\|^2)}$$

$$u(x) = \frac{\psi(x)}{\rho(x)} = \frac{c(x)}{a(x)}$$

- We can learn 'a', 'b' and 'c' using $L(\theta)$ and thus a controller and density certificate!

Safe Stabilization with NCDFs

Safety is guaranteed via Control Barrier Functions(CBFs)

Theorem 4. Define the set $\mathcal{C} := \{x \in \mathbb{R}^n : h(x) > 0\} \subset S$, where $h \in C^1(\mathbb{R}^n, \mathbb{R})$. Suppose there exists a positive, density function ρ and $\psi \in C^\infty(\mathbb{R}^n, \mathbb{R})$ such that $f(x)\frac{\rho(x)}{\|x\|} + g(x)\frac{\psi(x)}{\|x\|}$ is integrable on $\{x \in \mathbb{R}^n : \|x\| \geq 1\}$ such that:

$$\begin{aligned}\nabla \cdot (\rho(x)f(x) + \psi(x)g(x)) &> 0 \quad \text{for almost all } x \\ \rho(x)(L_f h(x) + \alpha(h(x))) + \psi(x)L_g h(x) &\geq 0\end{aligned}$$

Then, there exists a static state feedback controller $u(x)$ such a solution $\phi(t, x)$ of $\dot{x} = f(x) + g(x)u(x)$, starting from almost all $x_0 \in \mathcal{C}$, satisfies $\lim_{t \rightarrow \infty} \phi(t, x) = 0$ and, for all $x \in \mathcal{C}$ and $t \geq 0$, $\phi(t, x) \in \mathcal{C}$.

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Reinforce the
safety property:

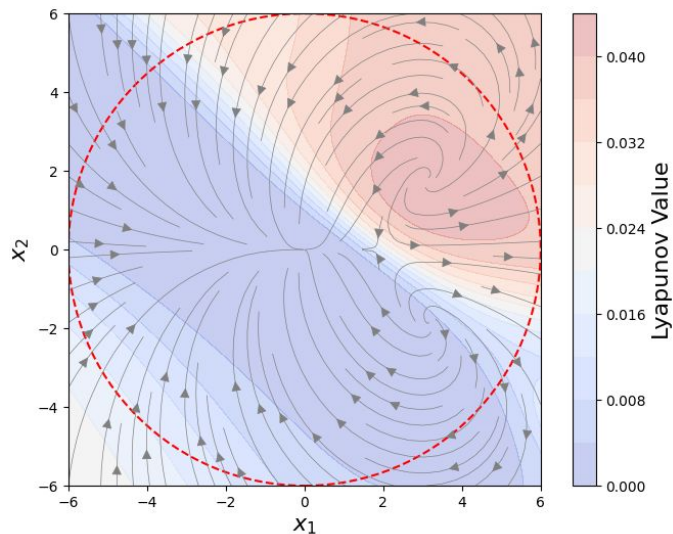
$$SL(\theta) = L(\theta) + \mathbb{E}[\max(0, -[\rho(x)(L_f h(x) + \alpha(h(x))) + \psi(x)L_g h(x)])]$$

Experiments: Lyapunov Functions fail with multiple equilibria

Consider the following system:

$$\dot{x}_1 = -2x_1 + x_1^2 - x_2^2 \quad \dot{x}_2 = -6x_2 + 2x_1x_2$$

It admits four equilibria: $(0,0), (2,0), (3, \pm\sqrt{3})$



Neural Lyapunov Method (Chang et al., 2019) fails!

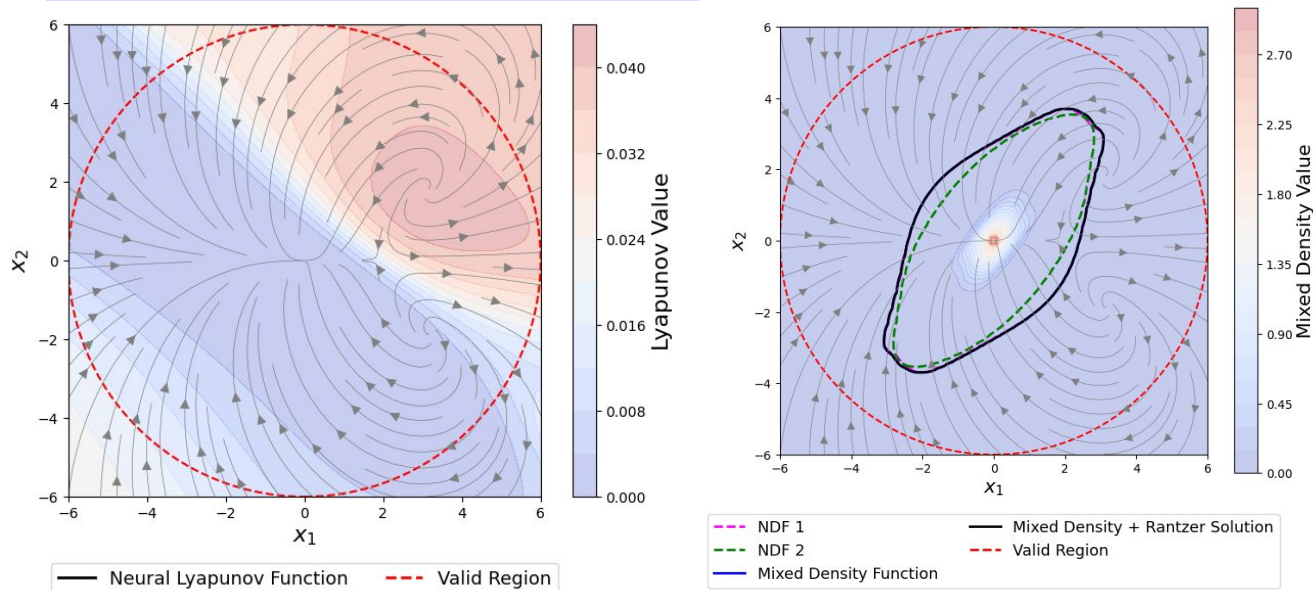
— Neural Lyapunov Function - - - Valid Region

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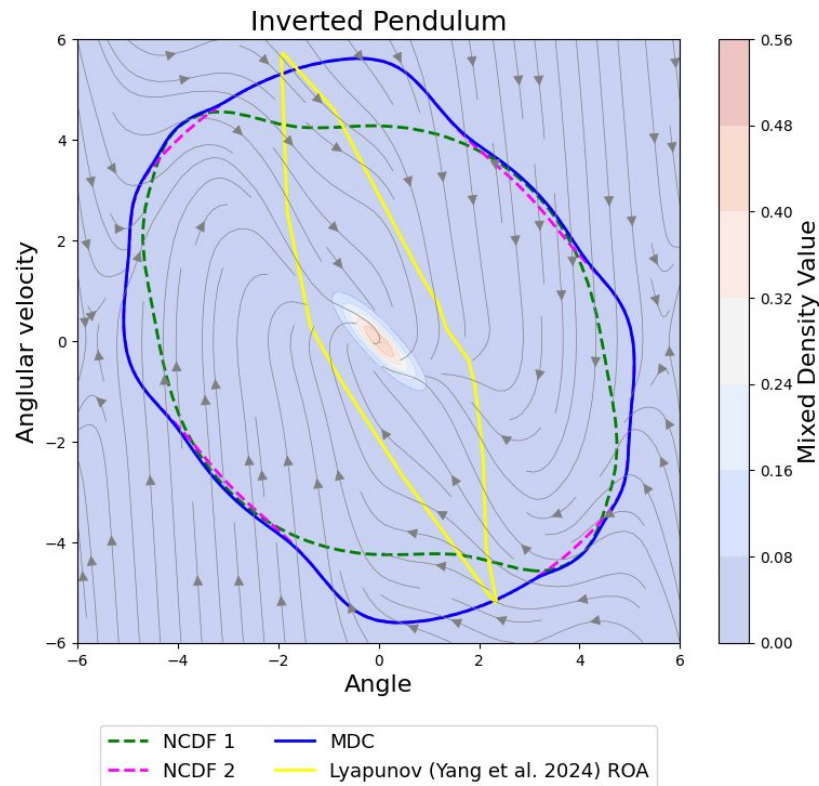


A valid Neural Density Function is learned which certifies a.e. stability!

Experiments: NCDFs and blending achieve SOTA Regions of Attraction!

Consider the problem of inverted pendulum:

$$\dot{\theta} = \dot{\theta}, \quad \ddot{\theta} = \frac{MgL \sin(\theta) - \mu \dot{\theta}}{ML^2} + \frac{u}{ML^2}$$



Thanks for your time!